Assignment 10

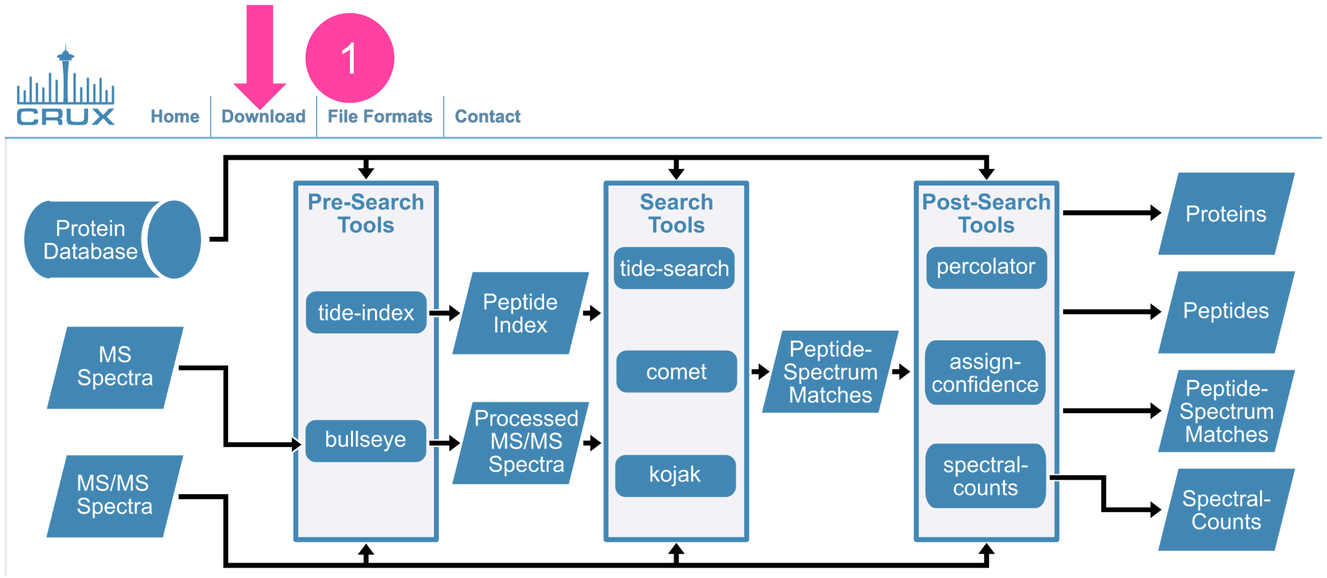
Proteogenomics

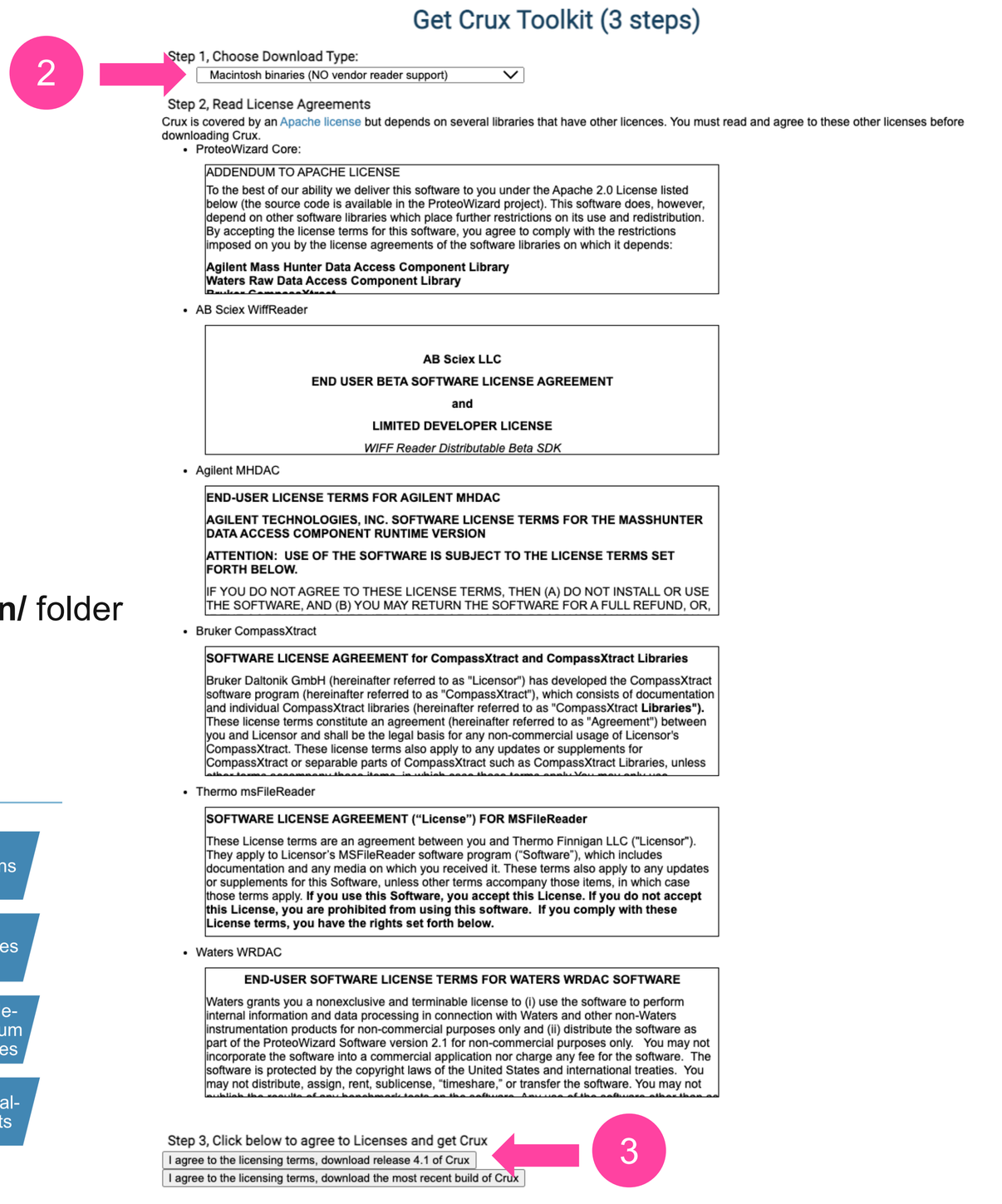
March 27 2026

Part 1 (two options)

Download Crux

1. Go to http://crux.ms/ and click Downloads
2. Select the download that matches your operating system
3. Agree to licensing terms and download release **4.3.2 of Crux**
4. Unzip the download
5. The program we'll be running from the command line is in the **bin/** folder





Part 1 (two options)

DO NOT activate conda environment

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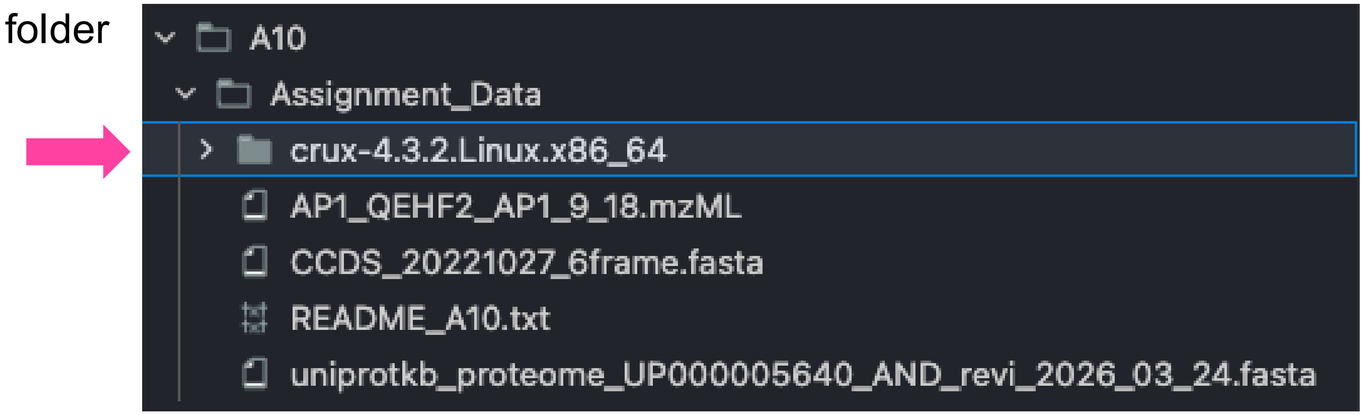
Use the Crux in Assignment_Data (Recommended)

1. Copy to your User folder

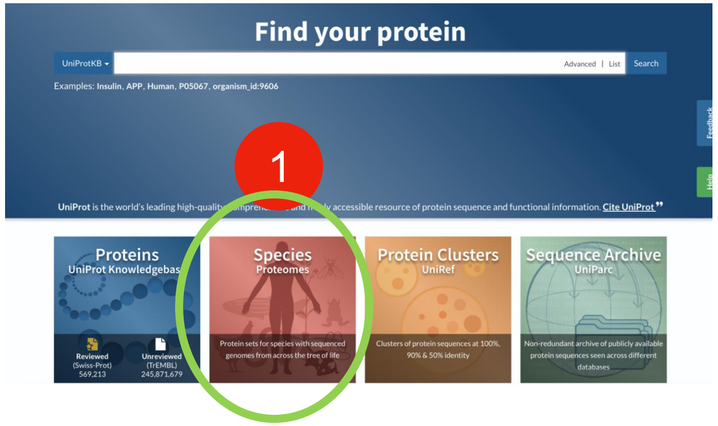


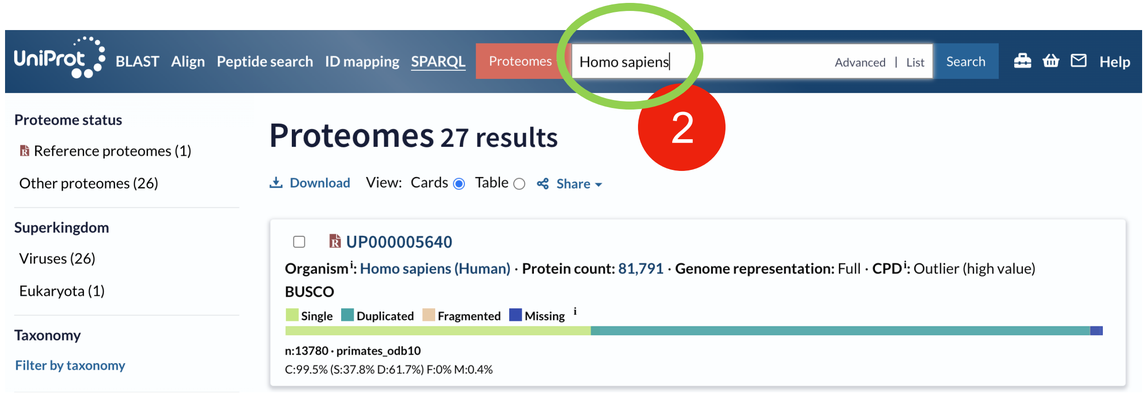
Question 0
(in README_A10.txt)
Please tell us the operation system you used

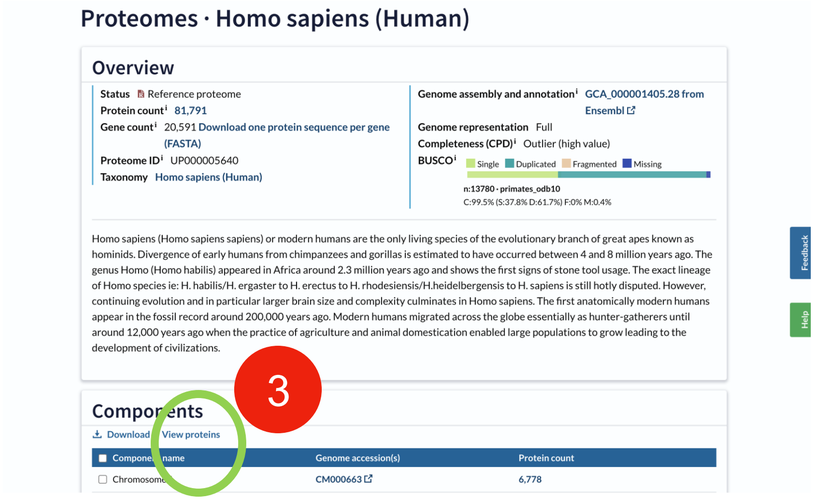
Part 2

Download Human Proteome

1. Go to https://www.uniprot.org/ and click Species Proteomes
2. Type “Homo sapiens” in the search bar and search
   - Select the UP000005640 proteome
3. Click View proteins (underneath Components)







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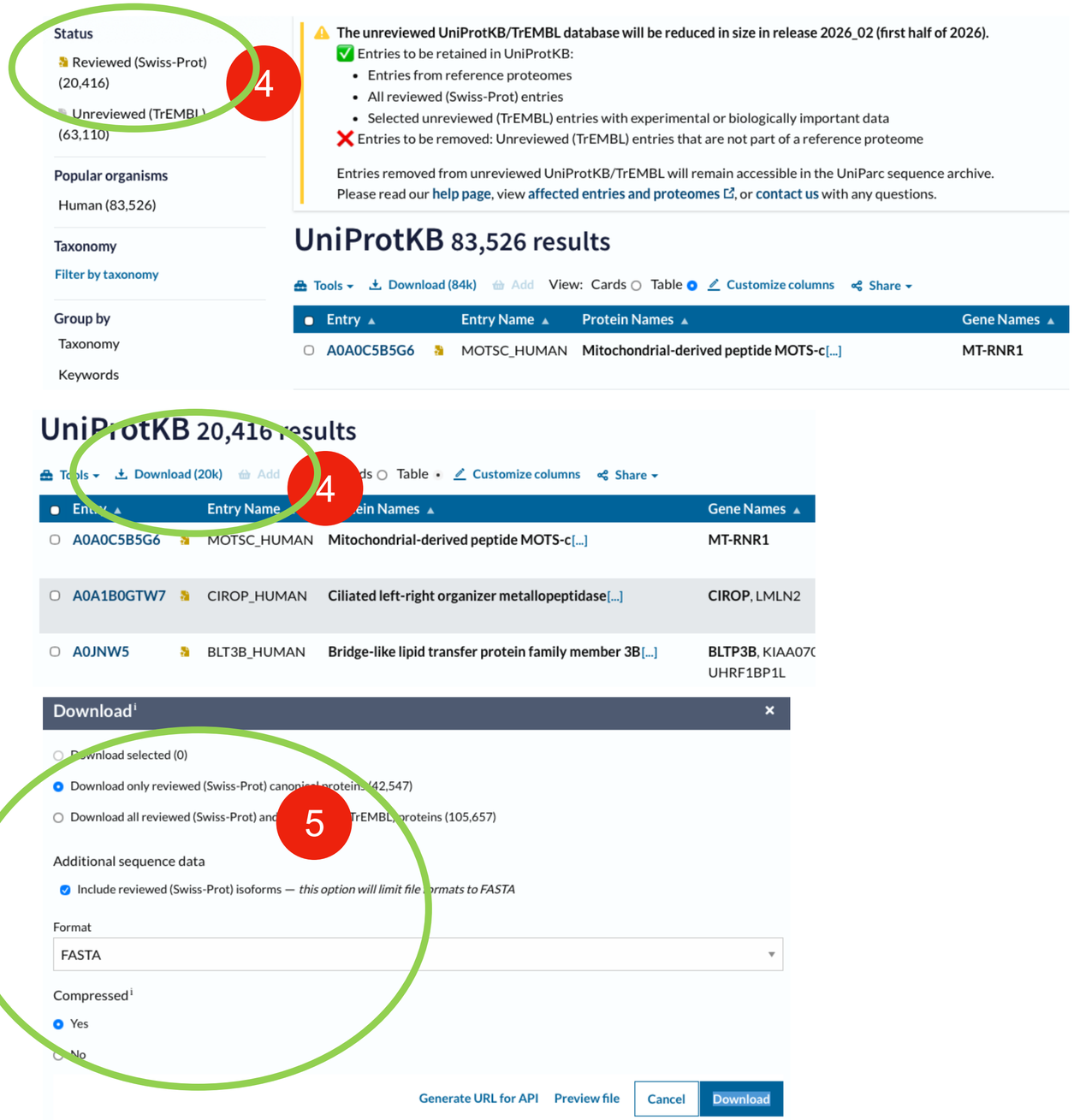
4. Select the “Reviewed (SwissProt)” Status on the top left
   - Click download
5. Select FASTA (Only reviewed canonical & isoforms) and uncompressed. Download. (**uniprotkb_proteome_UP000005640_AND_revi_2026_03_24** in Assignment_Data)



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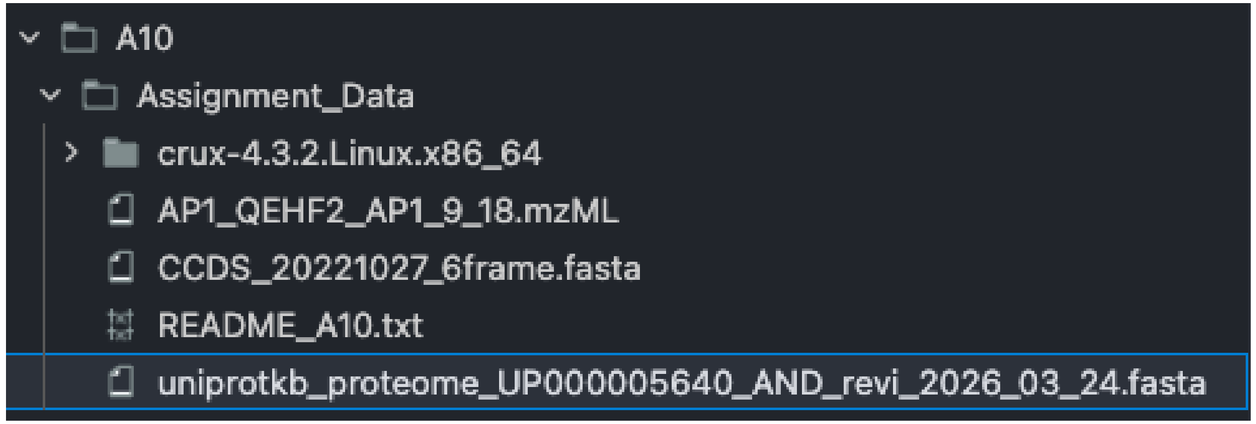

Part 3

Create a peptide database index for the human SwissProt proteome

1) Open up a terminal and create a peptide index for the human proteome.

2) Specify static TMT11 modifications on free amines (Lysines and peptide N-termini)

   --mods-spec K+229.162932 is a **static TMT11 modification on lysines**

   --nterm-peptide-mods-spec X+229.16293 is a static TMT11 mod on any residue on a peptide N-terminus.

   Use unimod.org (login as guest) to lookup masses of other PTMs if you're curious

3) Increase missed cleavages to 2

   --missed-cleavages 2 includes peptides with up to 2 missed cleavages by trypsin (e.g. PEPTIDE**RK**PEPTIDEK)

4) Use defaults for remaining settings

   - The digestion enzyme is set to **Trypsin** by default

   - The default **min and max peptide lengths are 6 and 50**, respectively.

5) This creates a directory called **human_sp_TMT11** that contains our peptide index (maps peptides to proteins, stores theoretical spectra)

Run the code on Github:

```
$ [PATH_TO_CRUX_BIN]/crux tide-index /
--overwrite T
--missed-cleavages 2
--mods-spec K+229.162932
--nterm-peptide-mods-spec X+229.16293
[PATH_TO_PROTEOME_FASTA]
human_swissprot_TMT11
```

Warning:

Because of some very subtle differences in hyphens in Word/ PDF vs your terminal, copy pasting these commands may sometime result in errors related to the number of arguments.

If that happens, just go through your code while on the command line, and delete and re-type all of the hyphens ("-")

Part 3

Create a peptide database index for the human SwissProt proteome

6) Explore database size change with variable modifications

`--mods-spec K+229.162932, 3M+15.994915` add up to 3 **oxidized methionine** per peptide

`--mods-spec K+229.162932, 3M+15.994915, 3STY+79.96633` further add up to 3 **phosphorylation** per peptide

```
INFO: Ignoring 17 peptide sequences containing unrecognized characters.
INFO: Generated 5763415 targets, including duplicates.
INFO: Writing peptides
INFO: Wrote 1000000 unique target peptides
INFO: Wrote 2000000 unique target peptides
INFO: Skipped 3088633 duplicate targets.
INFO: Generated 2674782 unique target peptides.
INFO: Computing modified peptides...
INFO: Wrote 10000000 modified target peptides to temp files
INFO: Wrote 20000000 modified target peptides to temp files
INFO: Wrote 30000000 modified target peptides to temp files
INFO: Wrote 40000000 modified target peptides to temp files
INFO: Wrote 50000000 modified target peptides to temp files
INFO: Wrote 60000000 modified target peptides to temp files
INFO: Wrote 70000000 modified target peptides to temp files
INFO: Wrote 80000000 modified target peptides to temp files
INFO: Wrote 90000000 modified target peptides to temp files
INFO: Wrote 100000000 modified target peptides to temp files
INFO: Wrote 110000000 modified target peptides to temp files
INFO: Wrote 120000000 modified target peptides to temp files
INFO: Wrote 130000000 modified target peptides to temp files
INFO: Created 130873363 modified and unmodified target peptides.
INFO: Generating 1 decoy(s) per target peptide
INFO: Wrote 10000000 target and their corresponding decoy peptides
INFO: Wrote 20000000 target and their corresponding decoy peptides
INFO: Wrote 30000000 target and their corresponding decoy peptides
INFO: Wrote 40000000 target and their corresponding decoy peptides
INFO: Wrote 50000000 target and their corresponding decoy peptides
INFO: Wrote 60000000 target and their corresponding decoy peptides
INFO: Wrote 70000000 target and their corresponding decoy peptides
INFO: Wrote 80000000 target and their corresponding decoy peptides
INFO: Wrote 90000000 target and their corresponding decoy peptides
INFO: Wrote 100000000 target and their corresponding decoy peptides
INFO: Wrote 110000000 target and their corresponding decoy peptides
INFO: Wrote 120000000 target and their corresponding decoy peptides
INFO: Wrote 130000000 target and their corresponding decoy peptides
INFO: Failed to generate decoys for 1590 low complexity peptides.
INFO: Generated 130873363 target peptides.
INFO: Generated 130868395 decoy peptides.
INFO: Generated 261741758 peptides in total.
INFO: Elapsed time: 778 s
INFO: Finished crux tide-index.
INFO: Return Code:0
```

For Question 1 (in README_A10.txt)

```
INFO: Generated [NUMBER] peptides in total.
```

Question 2 & 3 (in README_A10.txt)

Part 4

Download raw mass spectrometry data from CPTAC's lung adenocarcinoma study.

- Go to https://proteomic.datacommons.cancer.gov/pdc/
- Click **Lung Adenocarcinoma**
- Find the row that says **APOLLO-LUAD** Proteome. Click on the "**360**" under the mzML header.
  - You can use the .raw files as well, but Crux only provides support to read .raw files on windows.
- Download the **AP1_QEHF2_AP1_9_18.mzML.gz** file (already in Assignment_Data)
  - In the interest of time, we'll only download 1 file, but in reality each sample is divided into multiple fractions (usually 25) and often analyzed in two technical replicates.
- Unzip the file and stick into your assignment folder

PDC Study ID	Study	Project	Program	Disease Type	Primary Site	Analytical Fraction	Experiment Type	Cases #	Available files fo		
									Raw	mzML	Metadata
			APOLLO								
PDC000435	APOLLO LUAD - Phosphoproteome - TiO2	APOLLO1	Applied Proteogenomics Organizational Learning and Outcomes - APOLLO	Lung Adenocarcinoma	Bronchus and lung	Phosphoproteome	TMT11	100	120	120	1
PDC000434	APOLLO LUAD - Proteome	APOLLO1	Applied Proteogenomics Organizational Learning and Outcomes - APOLLO	Lung Adenocarcinoma	Bronchus and lung	Proteome	TMT11	100	360	360	1

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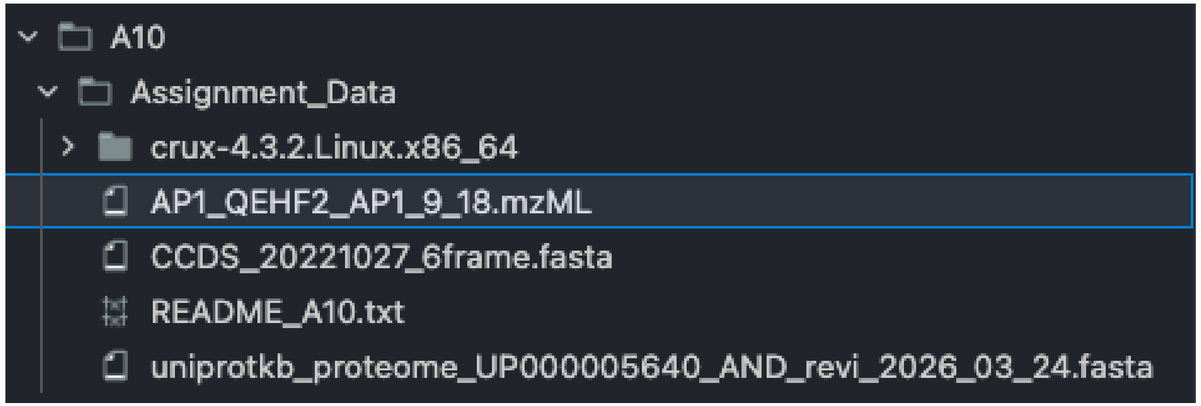

Part 5

Database Search with SwissProt proteome

- Use the terminal to run the following command:
- crux pipeline **AP1_QEHF2_AP1_9_18.mzML** **human_sp_TMT11/**
- This will run a database search (Tide/Sequest), a peptide inference method (Percolator), and false discovery rate estimation.
- The peptide identifications will be in crux-output/percolator.target.peptides.txt

```
denisgoldfarb@3225-AL-05002 Assignment10 % ../crux-4.1.Darwin.x86_64/bin/crux pipeline AP1_QEHF2_AP1_9_18.mzML human_sp_TMT11
WARNING: The output directory 'crux-output' already exists.
Existing files will not be overwritten.
INFO: CPU: 3225-AL-05002
INFO: Crux version: 4.1-e4b2f4d2-2021-10-13
INFO: Sun Mar 26 11:50:58 CDT 2023
INFO: Beginning pipeline.
INFO: Running pipeline with the following steps:
INFO: --> tide-search
INFO: --> percolator
INFO: Running tide-search...
INFO: Search will be run with the following files against database 'human_sp_TMT11':
INFO: --> AP1_QEHF2_AP1_9_18.mzML
INFO: Running tide-search...
INFO: Number of Threads: 1
INFO: Reading index human_sp_TMT11
INFO: Read 20591 target proteins
INFO: Converting AP1_QEHF2_AP1_9_18.mzML to spectrumrecords format
INFO: Elapsed time starting conversion: 1.68 s
INFO: Converting ms_level 2 ...
INFO: Reading spectrum file crux-output/AP1_QEHF2_AP1_9_18.mzML.spectrumrecords.tmp.
INFO: Read 34499 spectra.
INFO: Starting search.
INFO: 1000 spectrum-charge combinations searched, 3% complete
INFO: 2000 spectrum-charge combinations searched, 6% complete
INFO: 3000 spectrum-charge combinations searched, 9% complete
INFO: 4000 spectrum-charge combinations searched, 12% complete
INFO: 5000 spectrum-charge combinations searched, 14% complete
```

```
INFO:  0.1305    0.2136  0.1209 Charge3
INFO:  0.2138    0.2561  0.2969 Charge4
INFO:  0.0000    0.0000  0.0000 enzN
INFO:  0.0000    0.0000  0.0000 enzC
INFO: -0.2957   -0.6258 -0.5574 enzInt
INFO:  0.0247    0.0047 -0.0381 lnNumDSP
INFO:  0.1342    0.1800  0.1874 dM
INFO:  0.3183    0.2151  0.1710 absdM
INFO: -3.6579   -6.1172 -5.4837 m0
INFO: Found 5549 test set PSMs with q<0.01.
INFO: Selected best-scoring PSM per scan+expMass (target-decoy competition): 20860 target PSMs and 13160 decoy PSMs.
INFO: Multiple instantiations of Normalizer
INFO: Multiple instantiations of Normalizer
INFO: Multiple instantiations of Normalizer
INFO: Tossing out "redundant" PSMs keeping only the best scoring PSM for each unique peptide.
INFO: Calculating q values.
INFO: Final list yields 3172 target peptides with q<0.01.
INFO: Calculating posterior error probabilities (PEPs).
INFO: Processing took 45.0715 cpu seconds or 45 seconds wall clock time.
INFO: Multiple instantiations of Normalizer
INFO: Multiple instantiations of Normalizer
INFO: Elapsed time: 192 s
INFO: Finished crux pipeline.
INFO: Return Code:0
```

For Question 5 (in README_A10.txt)

```
INFO: Final list yields [NUMBER] target peptides with q<0.01.
```

Part 6

Database Search with 6-frame translation

- Download the 6-frame translated FASTA file from box:
  https://wustl.box.com/s/aduuzpxud4xzvabncz51o9tmx6btf6be
  (you can use **CCDS_20221027_6frame.fasta** in Assignment_Data)

- Create a peptide index of your 6-frame translated database ➔ **Question 4** (in README_A10.txt)

- Perform a database search on the same file ➔ **Question 5** (in README_A10.txt)

- Identify out-of-frame peptides with q-value less than 1% and blast-p them against human proteome to find potential in-frame explanations for them. Write out whether they had an in-frame explanation in the README file.
  ➔ **Question 6** (in README_A10.txt)

  - Does not matter how you identify in-frame peptides- use python, R, excel, whatever you please

  - For blast-p use human Swissprot and the rest can be default settings.

    - Pro-tip: look into submitting multiple sequences in a FASTA format

Summary

- Entire homework uses 2 commands with this structure
  - Command to index reference proteomes
    - crux_folder/bin/crux tide-index [OPTIONAL_ARGUMENTS] [PATH_TO_PROTEOME] [NAME_OF_OUTPUT_FOLDER]
  - Command to do search database of reference proteomes
    - crux_folder/bin/crux pipeline [OPTIONAL_ARGUMENTS] [PATH_TO_MS_DATA] [INDEXED_REFERENCE_PROTEOME]
- Keep in mind
  - The program overwrites its log files, so we suggest looking at your outputs from every command, and moving/copying/renaming the ones you need to submit.
  - Bulk of work for this homework is answering README questions, please do not leave it till the last minute

What to turn in:

- README_A10.txt file with answers to each of the questions and the commands as they were in your terminal or command line to generate the answers
- 4 tide-index.log.txt files from your indexing
  - human_swissprot_tide-index.log.txt
  - human_swissprot_oxM_tide-index.log.txt
  - human_swissprot_oxM_pSTY_tide-index.log.txt
  - human_CCDS_6frame _tide-index.log.txt
- 2 percolator.target.peptides.txt files from each of your database searches
  - human_swissprot_percolator.target.peptides.txt file
  - human_CCDS_6frame _percolator.target.peptides.txt file
- 2 pipeline.log.txt files from each of your database searches
  - human_swissprot_pipeline.log.txt file
  - human_CCDS_6frame_ pipeline.log.txt file